

# PHB 228 Homework 2

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## Part 1

### 1.1

```
library(dplyr)
```

Warning: package 'dplyr' was built under R version 4.5.2

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(car)
```

Warning: package 'car' was built under R version 4.5.2

Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:dplyr':

recode

```
mtcars <- as_tibble(mtcars)

# a)
model1 <- lm(mpg ~ wt + hp, data = mtcars)
summary(model1)
```

Call:

```
lm(formula = mpg ~ wt + hp, data = mtcars)
```

Residuals:

| Min    | 1Q     | Median | 3Q    | Max   |
|--------|--------|--------|-------|-------|
| -3.941 | -1.600 | -0.182 | 1.050 | 5.854 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | 37.22727 | 1.59879    | 23.285  | < 2e-16 ***  |
| wt          | -3.87783 | 0.63273    | -6.129  | 1.12e-06 *** |
| hp          | -0.03177 | 0.00903    | -3.519  | 0.00145 **   |

---

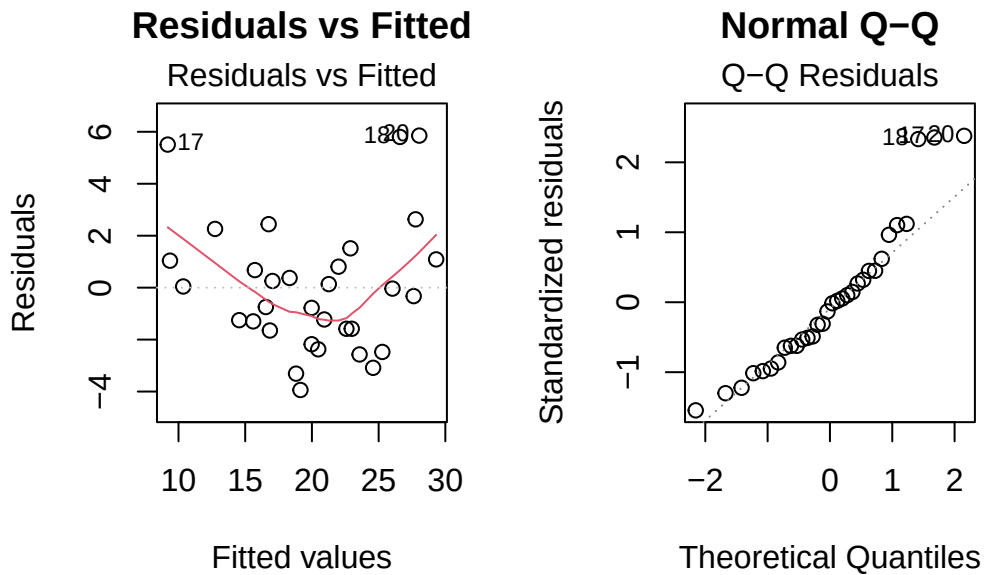
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.593 on 29 degrees of freedom

Multiple R-squared: 0.8268, Adjusted R-squared: 0.8148

F-statistic: 69.21 on 2 and 29 DF, p-value: 9.109e-12

```
# b)
par(mfrow = c(1, 2))
plot(model1, which = 1, main = "Residuals vs Fitted")
plot(model1, which = 2, main = "Normal Q-Q")
```



```
# c)
vif(model1)
```

```
      wt      hp
1.766625 1.766625
```

```
# d)
model2 <- lm(mpg ~ wt * hp, data = mtcars)
anova(model1, model2)
```

Analysis of Variance Table

Model 1: mpg ~ wt + hp

Model 2: mpg ~ wt \* hp

|   | Res.Df | RSS    | Df | Sum of Sq | F      | Pr(>F)        |
|---|--------|--------|----|-----------|--------|---------------|
| 1 | 29     | 195.05 |    |           |        |               |
| 2 | 28     | 129.76 | 1  | 65.286    | 14.088 | 0.0008108 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

1.1 a: Both wt and hp are highly significant ( $p < 0.001$ ). Holding horsepower constant, expected fuel efficiency decreases by 3.88 mpg for every 1000 lbs increase in weight. Holding

weight constant, expected fuel efficiency decreases by 0.032 mpg for every additional unit of horsepower.

1.1 b: The residuals appear reasonably scattered around zero without strong systematic structure, supporting the homoscedasticity assumption, although a few observations show moderate deviation. The Normal Q-Q plot approximately follows the diagonal, suggesting that the residuals are reasonably close to normally distributed with slight deviations in the tails.

1.1 c: `car::vif()` yields 1.766 for both variables, well below the standard threshold of 5 or 10, indicating no severe multicollinearity.

1.1 d: The ANOVA comparison yields  $F = 14.088$  and  $p < 0.001$ , indicating the interaction model provides a significantly better fit. The positive interaction term ( $wt:hp = 0.0278$ ) implies that the negative impact of horsepower on fuel efficiency is attenuated for heavier vehicles. Model 2 is preferred.

## 1.2

```
library(MASS)
```

Attaching package: 'MASS'

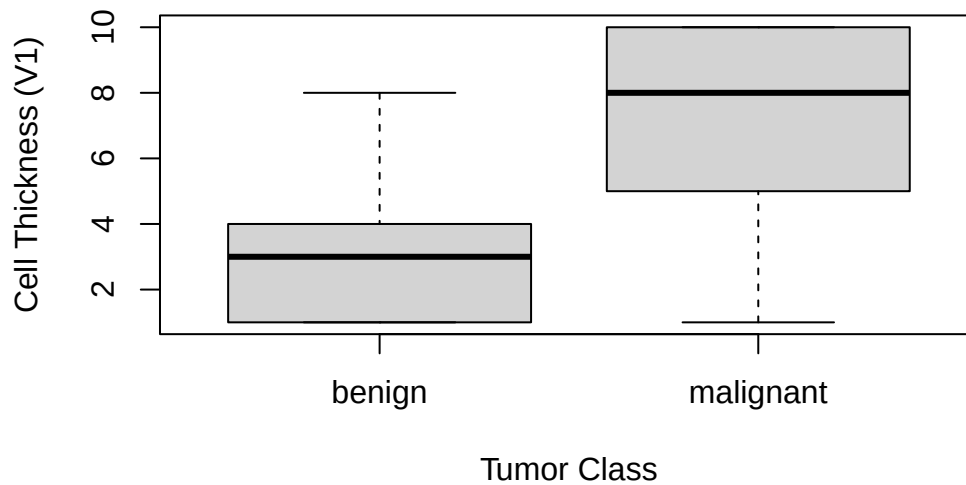
The following object is masked from 'package:dplyr':

```
select
```

```
data(biopsy, package = "MASS")
biopsy <- na.omit(biopsy)
biopsy$class <- factor(biopsy$class, levels = c("benign", "malignant"))

# a)
boxplot(V1 ~ class, data = biopsy,
        ylab = "Cell Thickness (V1)", xlab = "Tumor Class",
        main = "Cell Thickness by Tumor Classification")
```

## Cell Thickness by Tumor Classification



```
# b)
logit_model <- glm(class ~ V1 + V2 + V3, data = biopsy, family = binomial())
summary(logit_model)
```

Call:

```
glm(formula = class ~ V1 + V2 + V3, family = binomial(), data = biopsy)
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | -7.7210  | 0.6969     | -11.079 | < 2e-16 ***  |
| V1          | 0.5918   | 0.1030     | 5.746   | 9.14e-09 *** |
| V2          | 0.6390   | 0.1704     | 3.751   | 0.000176 *** |
| V3          | 0.7240   | 0.1661     | 4.358   | 1.31e-05 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 884.35 on 682 degrees of freedom  
 Residual deviance: 176.50 on 679 degrees of freedom  
 AIC: 184.5

Number of Fisher Scoring iterations: 7

```
exp(coef(logit_model))
```

|  | (Intercept)  | V1           | V2           | V3           |
|--|--------------|--------------|--------------|--------------|
|  | 0.0004433959 | 1.8073247714 | 1.8946201232 | 2.0627331079 |

```
# c)
biopsy_pred <- biopsy %>%
  mutate(pred_prob = predict(logit_model, type = 'response'),
         pred_class = factor(ifelse(pred_prob > 0.5, 'malignant', 'benign'),
                             levels = c('benign', 'malignant')))

confusion_mat <- table(Observed = biopsy_pred$class, Predicted = biopsy_pred$pred_class)
print(confusion_mat)
```

|           | Predicted |           |
|-----------|-----------|-----------|
| Observed  | benign    | malignant |
| benign    | 430       | 14        |
| malignant | 20        | 219       |

```
accuracy <- sum(diag(confusion_mat)) / sum(confusion_mat)
sensitivity <- confusion_mat['malignant', 'malignant'] / sum(confusion_mat['malignant', ])
specificity <- confusion_mat['benign', 'benign'] / sum(confusion_mat['benign', ])

# d)
new_prob <- predict(logit_model, newdata = data.frame(V1 = 5, V2 = 4, V3 = 4), type = 'response')
```

## 1.2 a

The boxplot reveals that malignant tumors have a substantially higher median cell thickness (V1) than benign ones. The lack of interquartile overlap suggests V1 is a strong discriminator.

## 1.2 b

All three predictors are highly significant. The odds ratios are approximately 1.81 for V1, 1.89 for V2, and 2.06 for V3, meaning each unit increase multiplies the odds of malignancy by the respective ratio.

## 1.2 c

The model achieves an accuracy of 0.950, a sensitivity of 0.916, and a specificity of 0.968.

## 1.2 d

The predicted probability of malignancy for a new observation with (V1=5, V2=4, V3=4) is approximately 0.666.

## Part 2

```
library(lme4)
```

Loading required package: Matrix

```
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.5.2

```
# a)
set.seed(123)
n_participants <- 20
n_items <- 15
n_conditions <- 2

rt_data <- expand.grid(
  participant = factor(1:n_participants),
  item = factor(1:n_items),
  condition = factor(c("control", "treatment"))
)

subject_intercept <- rnorm(n_participants, mean = 0, sd = 50)
item_intercept <- rnorm(n_items, mean = 0, sd = 20)

rt_data$reaction_time <- 500 +
  ifelse(rt_data$condition == "treatment", -30, 0) +
  subject_intercept[rt_data$participant] +
```

```

    item_intercept[rt_data$item] +
    rnorm(nrow(rt_data), mean = 0, sd = 30)

# b)
fixed_model <- lm(reaction_time ~ condition, data = rt_data)

randint_participant <- lmer(reaction_time ~ condition + (1 | participant), data = rt_data, REML = FALSE)

randint_both <- lmer(reaction_time ~ condition + (1 | participant) + (1 | item), data = rt_data, REML = FALSE)

aic_vals <- c(AIC(fixed_model), AIC(randint_participant), AIC(randint_both))
names(aic_vals) <- c('fixed', 'participant', 'participant_item')
print(aic_vals)

```

| fixed    | participant | participant_item |
|----------|-------------|------------------|
| 6604.758 | 6060.940    | 5903.881         |

```

# c)
print(summary(randint_both)$coefficients)

```

|                    | Estimate  | Std. Error | t value   |
|--------------------|-----------|------------|-----------|
| (Intercept)        | 506.26504 | 11.928474  | 42.44173  |
| conditiontreatment | -32.11015 | 2.412167   | -13.31174 |

```

variance_components <- as.data.frame(VarCorr(randint_both))
participant_var <- variance_components$vcov[variance_components$grp == 'participant']
item_var <- variance_components$vcov[variance_components$grp == 'item']
residual_var <- sigma(randint_both)^2

icc_participant <- participant_var / (participant_var + item_var + residual_var)
cat("Participant-level ICC:", icc_participant, "\n")

```

Participant-level ICC: 0.6497365

```

# d)
rand_slope <- lmer(
  reaction_time ~ condition + (condition | participant) + (1 | item),
  data = rt_data,
  REML = FALSE
)
print(anova(randint_both, rand_slope))

```



Data: rt\_data

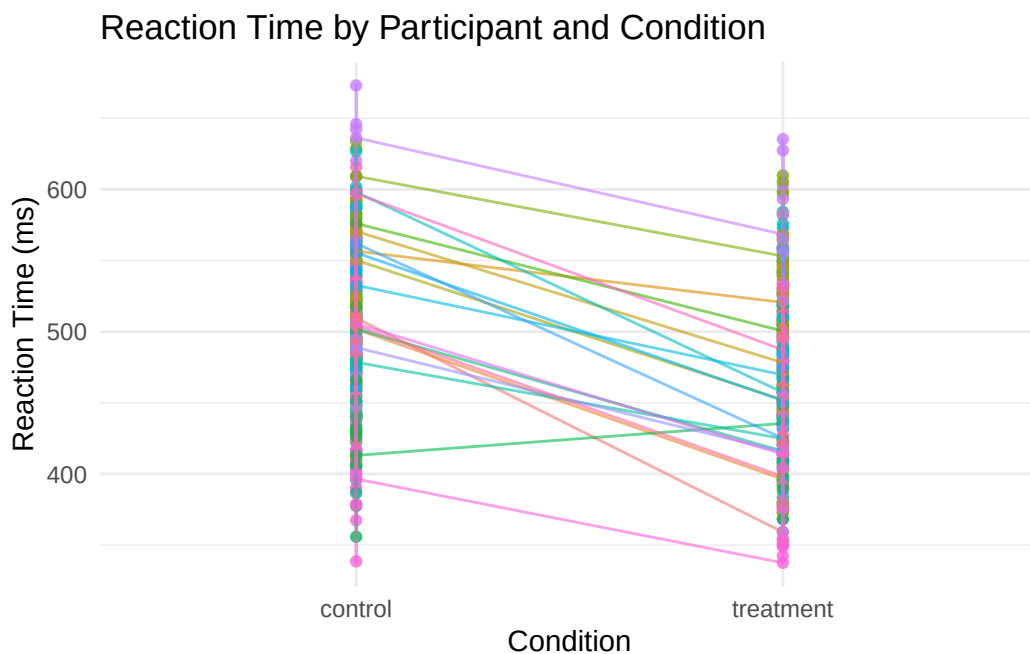
Models:

randint\_both: reaction\_time ~ condition + (1 | participant) + (1 | item)

rand\_slope: reaction\_time ~ condition + (condition | participant) + (1 | item)

|              | npar | AIC    | BIC    | logLik  | -2*log(L) | Chisq  | Df | Pr(>Chisq) |
|--------------|------|--------|--------|---------|-----------|--------|----|------------|
| randint_both | 5    | 5903.9 | 5925.9 | -2946.9 | 5893.9    |        |    |            |
| rand_slope   | 7    | 5906.1 | 5936.9 | -2946.1 | 5892.1    | 1.7335 | 2  | 0.4203     |

```
# e)
ggplot(rt_data, aes(x = condition, y = reaction_time, group = participant, color = participant)) +
  geom_point(size = 1.5, alpha = 0.8) +
  geom_line(alpha = 0.6) +
  labs(
    title = 'Reaction Time by Participant and Condition',
    x = 'Condition',
    y = 'Reaction Time (ms)'
  ) +
  theme_minimal() +
  theme(legend.position = 'none')
```



## 2 b

The random-intercept model for both participants and items yields the lowest AIC (5903.88) and is therefore preferred.

## 2 c

The fixed effect estimate for treatment is approximately -32.11, indicating treatment reduces average reaction time by about 32 ms. The participant-level ICC is 0.6497. This indicates that roughly 65% of the total variance in reaction time is attributable to between-participant baseline differences.

## 2 d

The random-slope model does not provide a statistically significant improvement over the random-intercept model ( $p = 0.4203$ ).

## Part 3

```
library(ggplot2)
library(dplyr)
library(palmerpenguins)
```

Attaching package: 'palmerpenguins'

The following objects are masked from 'package:datasets':

```
penguins, penguins_raw
```

```
penguins_clean <- penguins %>%
  filter(!is.na(bill_length_mm), !is.na(flipper_length_mm),
         !is.na(body_mass_g), !is.na(sex),
         !is.na(species), !is.na(island))

theme_penguin <- function() {
  theme_classic() +
```

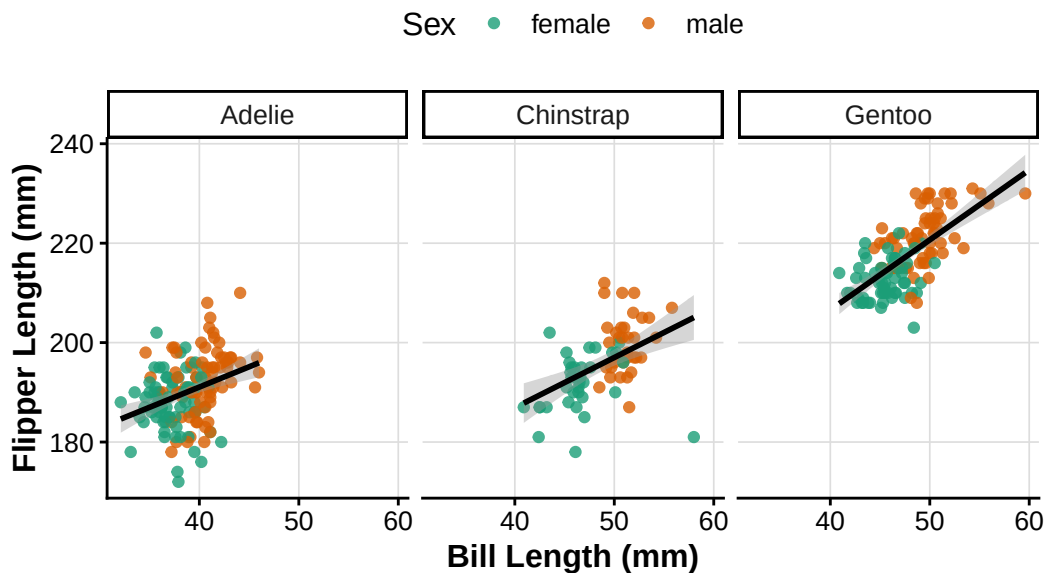
```

theme(
  text = element_text(size = 12, family = "sans"),
  plot.title = element_text(face = "bold", size = 16, hjust = 0.5),
  plot.subtitle = element_text(size = 11, color = "#555555"),
  axis.title = element_text(face = "bold"),
  panel.grid.major = element_line(color = "#dddddd", linewidth = 0.3),
  panel.grid.minor = element_blank(),
  legend.position = "top",
  legend.background = element_rect(fill = "transparent"),
  legend.key = element_rect(fill = "transparent", color = NA)
)
}
p1 <- ggplot(penguins_clean, aes(x = bill_length_mm, y = flipper_length_mm, color = sex)) +
  geom_point(alpha = 0.8) +
  geom_smooth(method = "lm", formula = y ~ x, se = TRUE, color = "black", linetype = "solid") +
  facet_wrap(~ species) +
  scale_color_brewer(palette = "Dark2") +
  labs(title = "Bill Length vs Flipper Length by Species",
       x = "Bill Length (mm)", y = "Flipper Length (mm)", color = "Sex") +
  theme_penguin()

p1

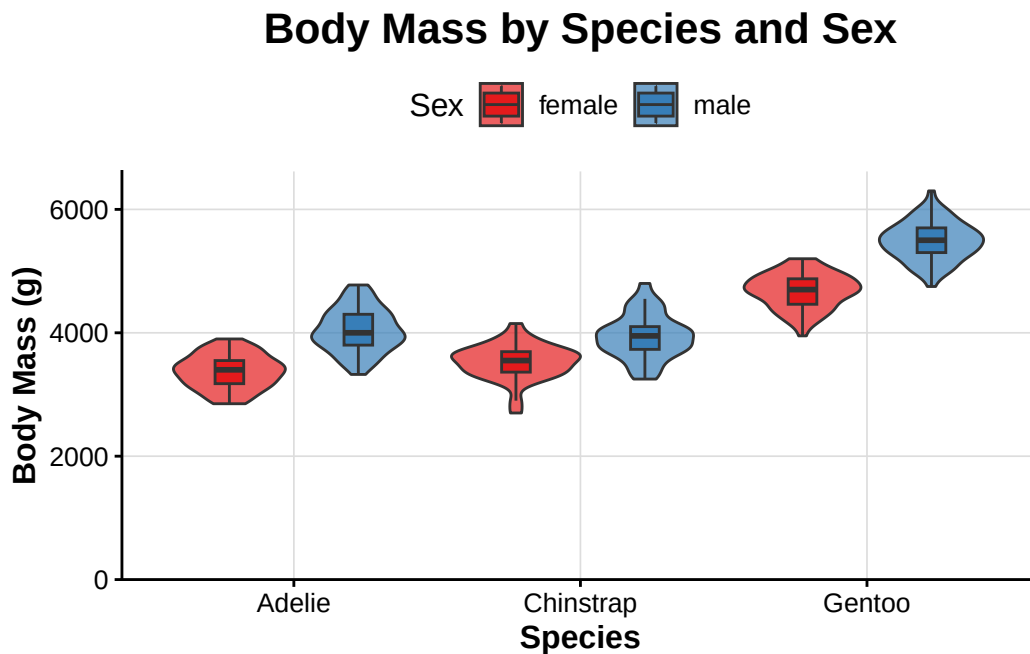
```

## Bill Length vs Flipper Length by Species



```
p2 <- ggplot(penguins_clean, aes(x = species, y = body_mass_g, fill = sex)) +
  geom_violin(position = position_dodge(width = 0.9), alpha = 0.7) +
  geom_boxplot(position = position_dodge(width = 0.9), width = 0.2, outlier.shape = NA) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.05)), limits = c(0, NA)) +
  scale_fill_brewer(palette = "Set1") +
  labs(title = "Body Mass by Species and Sex",
       x = "Species", y = "Body Mass (g)", fill = "Sex") +
  theme_penguin()
```

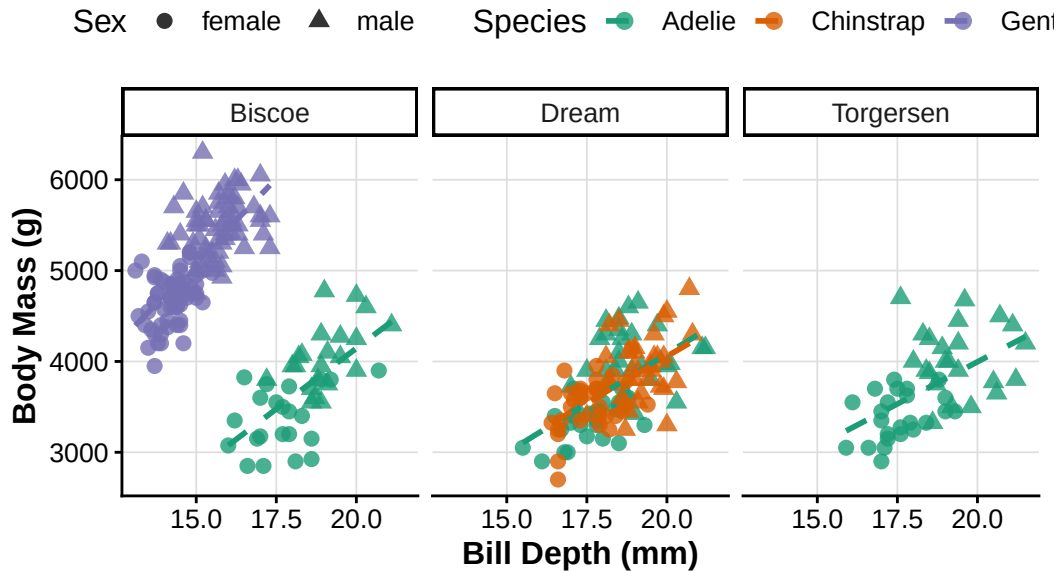
p2



```
p3 <- ggplot(penguins_clean, aes(x = bill_depth_mm, y = body_mass_g, color = species)) +
  geom_point(aes(shape = sex), size = 2.5, alpha = 0.8) +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE, linetype = "dashed") +
  facet_wrap(~ island) +
  scale_color_brewer(palette = "Dark2") +
  labs(title = "Body Mass vs Bill Depth by Species and Island",
       x = "Bill Depth (mm)", y = "Body Mass (g)",
       color = "Species", shape = "Sex") +
  theme_penguin()
```

p3

## Body Mass vs Bill Depth by Species and Island



The faceted scatter plot reveals positive relationships between bill depth and body mass within several penguin groups. Gentoo penguins generally exhibit larger body mass values than Adelie and Chinstrap penguins, while males tend to be heavier than females across species. Differences across islands also suggest potential environmental or ecological influences on penguin morphology. Overall, the visualization demonstrates that relationships between body mass and bill characteristics vary across biological subgroups, and aggregated patterns may obscure subgroup-specific trends.

## Part 4

```
library(shiny)
```

Warning: package 'shiny' was built under R version 4.5.2

```
library(ggplot2)
library(dplyr)

ui <- fluidPage(
  titlePanel('mtcars MPG Explorer'),
  sidebarLayout(
```

```

    sidebarPanel(
      selectInput('predictor', 'Predictor variable', choices = c('wt', 'hp', 'disp', 'drat')),
      sliderInput('cyl_filter', 'Cylinder range', min = min(mtcars$cyl), max = max(mtcars$cyl)),
      checkboxInput('show_line', 'Show regression line', value = TRUE)
    ),
    mainPanel(
      plotOutput('scatterPlot'),
      verbatimTextOutput('modelSummary')
    )
  )
)

server <- function(input, output) {

  filtered_data <- reactive({
    mtcars %>% filter(cyl >= input$cyl_filter[1], cyl <= input$cyl_filter[2])
  })

  fitted_model <- reactive({
    lm(as.formula(paste('mpg ~', input$predictor)), data = filtered_data())
  })

  output$scatterPlot <- renderPlot({
    p <- ggplot(filtered_data(), aes(x = .data[[input$predictor]], y = mpg)) +
      geom_point(color = '#2c3e50', size = 2) +
      labs(title = paste('mpg vs', input$predictor), x = input$predictor, y = 'mpg') +
      theme_minimal()

    if (input$show_line) {
      p <- p + geom_smooth(method = 'lm', se = FALSE, color = '#e74c3c')
    }
    p
  })

  output$modelSummary <- renderPrint({
    summary(fitted_model())
  })
}

shinyApp(ui = ui, server = server)

```

The app includes a cylinder filter and a toggle for the regression line. This allows users to

conduct conditional analysis by isolating hardware architectures, mitigating confounding while maintaining control over the visual overlay.

## Part 5

```
library(parallel)
library(microbenchmark)

physical_cores <- detectCores(logical = FALSE)
logical_cores <- detectCores(logical = TRUE)

safe_cores <- max(1, logical_cores - 1)

slow_mean <- function(x) {
  Sys.sleep(0.01)
  mean(x)
}

test_list <- replicate(100, rnorm(100), simplify = FALSE)

bench_res <- microbenchmark(
  sequential = lapply(test_list, slow_mean),
  parallel = mclapply(test_list, slow_mean, mc.cores = safe_cores),
  times = 5
)
```

Warning in microbenchmark(sequential = lapply(test\_list, slow\_mean), parallel = mclapply(test\_list, : less accurate nanosecond times to avoid potential integer overflows

```
print(bench_res)
```

```
Unit: milliseconds
      expr      min       lq      mean    median      uq      max  neval
sequential 1165.0229 1173.5367 1192.0027 1179.6411 1181.9917 1259.8213     5
  parallel  179.8573  180.4287  185.5521  183.4912  188.8023  195.1808     5
```

```

bootstrap_ci_sequential <- function(data, n_bootstrap = 1000, confidence = 0.95) {
  n <- length(data)
  boot_means <- numeric(n_bootstrap)
  for (i in seq_len(n_bootstrap)) {
    boot_means[i] <- mean(sample(data, n, replace = TRUE))
  }
  alpha <- 1 - confidence
  c(lower = quantile(boot_means, alpha / 2), upper = quantile(boot_means, 1 - alpha / 2))
}

bootstrap_ci_parallel <- function(data, n_bootstrap = 1000, confidence = 0.95) {
  n <- length(data)
  boot_means <- unlist(mclapply(seq_len(n_bootstrap), function(i) {
    mean(sample(data, n, replace = TRUE))
  }, mc.cores = safe_cores))
  alpha <- 1 - confidence
  c(lower = quantile(boot_means, alpha / 2), upper = quantile(boot_means, 1 - alpha / 2))
}

sizes <- c(500, 1000, 2000)

timing_results <- data.frame()

for (s in sizes) {

  seq_time <- system.time(
    bootstrap_ci_sequential(mtcars$mpg, n_bootstrap = s)
  )["elapsed"]

  par_time <- system.time(
    bootstrap_ci_parallel(mtcars$mpg, n_bootstrap = s)
  )["elapsed"]

  timing_results <- rbind(
    timing_results,
    data.frame(
      bootstrap_size = s,
      sequential = seq_time,
      parallel = par_time
    )
  )
}

```



```
print(timing_results)
```

|          | bootstrap_size | sequential | parallel |
|----------|----------------|------------|----------|
| elapsed  | 500            | 0.010      | 0.036    |
| elapsed1 | 1000           | 0.008      | 0.049    |
| elapsed2 | 2000           | 0.018      | 0.052    |

For relatively small bootstrap problems, parallel execution may not always outperform sequential execution because process management overhead can dominate the computational savings. However, as the number of bootstrap iterations increases, the relative efficiency of parallel processing generally improves. On Windows systems, `mclapply` has limited multicore support because forked processes are unavailable.